

signature Σ , $V = \{x, y\}$

$Val_x = \{0 \dots 9\}$ & $Val_y = \{\text{True}, \text{False}\}$

$F_x(v) = 0$, $F_y(v) = [\text{Proj}_x(v) > 5]$

M has $X \rightarrow Y$ because $X \propto Y$.

we can also say $F_y(x) = [x > 5]$

"Y is true exactly when $x > 5$ "

① $\gamma = \{\text{true}\} \in \text{Hard}_y$

② $I = v \mapsto [\text{Proj}_x(v) \leq 5] \in \text{Soft}_y$

Take the snapshot v & return true if $x \leq 5$

- soft intervention hands you a new rule for y & says use this instead

③ $\Pi = F_y \mapsto (\underbrace{v \mapsto \neg F_y(v)}_{\text{part 1}}) \in \text{Func}_y$
part 2

part 1: a rule that, given a snapshot v , computes whatever the old rule would have said & negates it.

part 2: take the old rule as input & build that negated rule from it.

• In summary: Π is the instruction

"make y do the opposite of whatever it currently does"

• $\mathbb{J} = \{F_x, F_y\} \mapsto \left\{ \underbrace{v \mapsto 6x \pm [\text{Proj}_y(v)]}_{F_x}, \underbrace{v \mapsto [\neg F_y(v)]}_{F_y} \right\} \in \text{Func}_v$

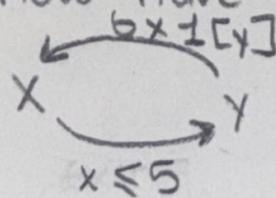
① new rule for x : $v \mapsto 6x \pm [\text{Proj}_y(v)]$

② new rule for y : $v \mapsto [\neg F_y(v)]$

"X becomes 6 if Y is true, 0 if Y is false"
shorthand: $F_x(y) = 6x \pm [y]$

→ negate y 's old rule, same as before

- we now have a cycle: $y \propto x$ & $x \propto y$



- Solve $(M_{\mathbb{J}}) = \emptyset$.

① $x \leq y$ ② $x \geq y$

② $x \leq 5$

$$y = 5 \rightarrow y = \text{false} (0)$$

$$y = x \rightarrow \text{contradiction!}$$

$$\left\{ \begin{array}{l} \text{solve } (M_j) = \emptyset; \end{array} \right.$$

$M \uparrow \text{ or } M \downarrow$: x remains $6 \times 4 [y]$
 $x < 5$: y becomes $7 (x \leq 5)$

$x = 1 \rightarrow y = \text{true}(1)$ ✓
 $x = 0 \rightarrow y = \text{false}(0)$ ✓

$$\left. \begin{aligned} & \text{solve} \\ & (H \text{ } \neg \text{ } H) \\ & = \{ \{6, \text{true}\}, \{0, \text{false}\} \} \end{aligned} \right\}$$

Definition 15:

A set of complicated interventions counts as an intervention algebra when structurally it behaves exactly like ordinary hard interventions

$(\Lambda, \oplus)$ is an intervention algebra if it is isomorphic to $(\Phi, \circ)$

- Λ = set of interventions
- $\oplus$ = How we combine those interventions
- Φ = set of normal hard interv.
- $\circ$ = doing one interv. after the other